The extended evidence-accumulation model for delayed reports

immediate

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1 The model

Time t runs from motion-stimulus onset. Motion of coherence c and direction $s \in \{-1, +1\}$ is presented for a duration D. Encoding time for the motion evidence is fixed across conditions and equal to L_{motion}, so encoded motion evidence is available to the decision process on $[L_{\text{motion}},\ L_{\text{motion}} + D]$.

1.1 Accumulation

The decision variable x and the motor-preparatory state y obey

$$\mathrm{d}x(t) = s\,k\,c\,\mathrm{d}t + \sigma_x\,\mathrm{d}W_x(t), \qquad \mathrm{d}y(t) = \lambda\,[x(t) - y(t)]\,\mathrm{d}t,$$

where W_x is a standard Wiener process and σ_x is taken as a fixed scaling parameter. Only kc/σ_x has meaning: the model is invariant under a common rescaling of x, y, θ, g, r and the maintenance noise, so one amplitude has to be declared, and σ_x is it.

1.2 Commitment

Maintenance begins at

$$\tau := \inf\{t \in [L_{\text{motion}}, t_{\text{offset}}] : |y(t)| \geq \theta\} \wedge t_{\text{offset}}, \qquad t_{\text{offset}} := D + L_{\text{motion}},$$

with $s_{\text{resp}} := \operatorname{sgn} y(\tau)$ and maintenance target

$$m := \begin{cases} s_{\text{resp}}\,\theta, & \text{if the threshold was reached,} \\ y(t_{\text{offset}}), & \text{otherwise,} \end{cases}$$

where θ is a threshold smaller than g. Note that $y(\tau) = m$ in both branches: if the threshold is reached, y hits θ continuously; if it is not, $\tau = t_{\text{offset}}$ and $m = y(t_{\text{offset}})$ by

definition. The maintained state therefore has no approach transient, and $m = s_{\text{resp}}|m|$ with $|m| = \min(|y(t_{\text{offset}})|, \theta)$ pathwise.

1.3 Maintenance

Let $t_{\text{RS}} := D + 3 + L_{\text{RS}}$ be the time at which the response signal is encoded. On $[\tau, t_{\text{RS}})$ the motor-preparatory state has two parts.

The first is a noisy maintenance of a baseline at the target m,

$$\mathrm{d}y(t) = \lambda\,[m + \varepsilon(t) - y(t)]\,\mathrm{d}t,$$

where

$$\mathrm{d}\varepsilon(t) = -\kappa\varepsilon(t)\,\mathrm{d}t + \sigma_\varepsilon\,\mathrm{d}W_\varepsilon(t), \quad \varepsilon(\tau) = 0,$$

is an Ornstein–Uhlenbeck process relaxing to zero at rate κ, with W_ε independent of W_x, and stationary standard deviation $\mathrm{SD}_\varepsilon = \sigma_\varepsilon/\sqrt{2\kappa}$. The initial condition and that standard deviation are not in conflict: ε starts at zero and its variance grows as $\mathrm{SD}_\varepsilon^2(1 - e^{-2\kappa\rho})$, so SD_ε describes the process only once the transient has passed, which takes a few multiples of $1/2\kappa$. Were $\kappa = 0$, that is, were the noise a Wiener process rather than an Ornstein–Uhlenbeck one, its standard deviation would grow without bound over the delay.

The second is temporal preparation, added optionally, and modelled as a linear ramp from zero at t_{offset} to β at t_{RS}, that is over $\Delta := t_{\text{RS}} - t_{\text{offset}} = 3 + L_{\text{RS}} - L_{\text{motion}}$, at rate $\alpha := \beta/\Delta$. The signal reaching the gate is

$$\tilde{y}(t) = y(t) + s_{\text{resp}}\,\beta\,\min\!\left(1, \frac{(t - t_{\text{offset}})^+}{\Delta}\right),$$

frozen at $s_{\text{resp}}\beta$ once $t > t_{\text{RS}}$, and the neural drive to the selected agonist is

$$z(t) = \max\bigl[s_{\text{resp}}\,\tilde{y}(t) - g,\, 0\bigr].$$

Model 1 and Model 2. Model 1 is the case $\beta = 0$, with no temporal preparation; Model 2 is $\beta > 0$.

1.4 Response stage

At t_{RS} the maintenance noise is discontinued and

$$\mathrm{d}q(t) = \mu_{\text{RS}}\mathrm{d}t + \sigma_q\mathrm{d}W_q(t),\ q(t_{\text{RS}}) = 0, \qquad \mathrm{d}y(t) = \lambda\,[\,y(t_{\text{RS}}) + s_{\text{resp}}\,q(t) - y(t)\,]\,\mathrm{d}t,$$

where W_q is a Wiener process independent of both W_x and W_ε. The response is triggered when z reaches r, and occurs L_{motor} later. An excursion of z above zero that returns to zero without reaching r is a partial activation.

2 Theoretical predictions

2.1 Behaviour

Behaviour here means what is available without electromyography: accuracy and response time, that is the first six rows of Table S1. Responses made before the signal are excluded from the analyses, and the model treats them as a nuisance rather than a dependent variable.

Accuracy rises with coherence and with duration (R1, R2). [1] The choice is fixed at commitment, as the sign of $y(\tau)$, and nothing after commitment can change it. Accuracy therefore reflects the first accumulation alone: it rises with coherence because the drift kc is larger relative to σ_x, and with duration because the accumulation runs longer before it is stopped. It is the one measure in the experiment that carries no information about anything downstream of commitment.

The duration benefit is confined to intermediate coherence (R3). At zero coherence there is no signal to accumulate, so more time cannot help; at fifty per cent the threshold θ is reached almost at once, so more time is not needed. The benefit of duration is therefore squeezed into the intermediate level, which is what the interaction reports. This is a floor-and-ceiling effect and not a mechanism: it requires only that θ be reachable within 0.3 s at high coherence and not at all at zero.

Response time falls with coherence and with duration (R4, R5). At the response signal the second accumulation starts at a distance $g - |m| - \beta$ below the gate, and its drift is the same in every condition. Coherence and duration can therefore act on response time only through that starting distance. Raising either raises the maintained level $|m|$ — higher coherence because the drift is larger, longer duration because accumulation is stopped later — which shortens the distance left to travel. Nothing downstream of the starting point depends on the stimulus.

The coherence effect on response time is larger after a short stimulus (R6). The maintained level is capped at θ: a trial that reaches the threshold is held there whatever its coherence. After a long stimulus most trials reach it at every coherence, so their starting points are compressed towards a common value and the coherence effect shrinks; after a short stimulus fewer do, and the starting points spread. The interaction therefore follows from the cap alone, with no further assumption. It is worth marking that this same argument, applied to the offset-locked rate, gives the opposite sign to what is observed there — see R19 below and Section 9.

2.2 During the delay

The four analysis windows are locked to the four trial events and do not coincide with the intervals they are named after, so this note calls them the *onset-locked*, *offset-locked*,

[1] Labels R1 to R27 index the results of Tables S1 and S2, reproduced with an added column in Appendix A.

signal-locked and *response-locked* windows; their bounds are given in Appendix A. The distinction matters most for the first, which contains 1.7 s of delay in the short-duration condition and none in the long one.

Partial activations are upcrossings of the gate by the maintained fluctuation. Appendix B derives their rate: writing $V := y - m$ for the fluctuation, σ_V and $\sigma_{\dot{V}}$ for its stationary standard deviations, $\omega := \sigma_{\dot{V}}/\sigma_V = \sqrt{\lambda\kappa}$ for its characteristic frequency, and

$$d(t) := g - |m| - \beta\,\frac{(t - t_{\text{offset}})^+}{\Delta}, \qquad a(t) := \frac{d(t)}{\sigma_V}, \qquad b := \frac{\alpha}{\sigma_V}$$

for the distance to the gate in absolute and in standardised units and for the ramp rate in standardised units, the rate is

$$\nu(t) = \varphi\big(a(t)\big)\,G, \qquad G := b\,\Phi\left(\frac{b}{\omega}\right) + \omega\,\varphi\left(\frac{b}{\omega}\right), \tag{1}$$

with G constant in time. Everything in this subsection follows from (1). The fluctuation is stationary throughout the analysis window, for the reason given with the slope below.

The duration effects in the onset-locked window separate the two models (R11, R14). The preparation ramp starts at t_{offset}, so it runs through three quarters of the short condition's window and none of the long condition's. Model 2 consequently predicts a higher rate and a steeper rise after a short stimulus, which is what R11 and R14 report. Model 1 has no ramp, its rate follows the level alone, and the level is higher after a long stimulus: it predicts the opposite sign for both. These are the only results outside the delay that separate the two models.

The rate rises through the onset-locked window, faster at higher coherence (R12, R13, R15). Higher coherence means a larger drift, hence a level that climbs faster and further, hence more crossings and a steeper rise. Both models predict this, since neither the drift nor the accumulation depends on the ramp.

The offset-locked rate rises with the maintained level (R16, R17). By (1) the rate depends on the level only through a, and φ decreases on $a > 0$, so anything that raises $|m|$ raises the rate. Higher coherence and longer duration both do, which gives the main effects of coherence and of duration on offset-locked rate. Neither depends on β: both models predict them.

The rate ramps if and only if there is temporal preparation, and the ramp decelerates (R18). With $a(t) = a_0 - bt$,

$$\frac{\mathrm{d}}{\mathrm{d}t}\ln\nu = b\,a(t), \qquad \frac{\mathrm{d}^2}{\mathrm{d}t^2}\ln\nu = -b^2, \tag{2}$$

so the log rate rises at a rate proportional to the current distance, and its second derivative is negative. If $\beta = 0$ then $b = 0$ and the rate is *exactly* constant. Both expressions are the stationary forms; the next paragraph shows that the fluctuation is stationary throughout the window.

A model without preparation cannot produce the offset-locked ramp (R18, R21). The constancy just stated holds once the maintained fluctuation is stationary, and a model with $\beta = 0$ might seem able to escape it: the fluctuation starts at $V(\tau) = 0$ and its variance builds up, which shrinks a and so raises the rate with no ramp at all. That escape is closed by two facts, and no calculation beyond arithmetic is needed.

The variance approaches its stationary value with time constant $1/2\kappa$, so a rate still climbing steeply at the end of a three-second delay requires $1/2\kappa$ of that order, that is $\kappa \lesssim 0.2$ s^{-1}. But with $\beta = 0$ equation (1) reduces to $\nu = (\omega/2\pi)e^{-a^2/2}$, so the rate can never exceed $\omega/2\pi = \sqrt{\lambda\kappa}/2\pi$ whatever the distance to the gate — and at $\kappa = 0.2$ that ceiling is 0.36 s^{-1}, or 0.43 once the transient is allowed for exactly. The observed rate at the end of the delay is 1.45 s^{-1}. A fluctuation slow enough to still be rising is far too slow to produce the rate; a fluctuation fast enough to produce the rate has been stationary since the first second, and then the rate is exactly flat. That second branch is also what licenses the stationary forms used above.[2]

Either way Model 1 predicts an offset-locked profile that is flat, or nearly so, at the end of the window. The observed profile over the last second alone runs 0.068, 0.074, 0.082, 0.085, 0.095, 0.098, 0.110, 0.119, 0.126 and 0.133 per bin, a log-rate slope of $+0.700 \pm 0.085$, $t(21) = 8.26$, $p = 5 \times 10^{-8}$. The discrimination between the two models rests on this, and on nothing else — no burst durations, no chronometry, no fitted parameters.

Cells with a higher rate must ramp more slowly (R20, R27). A higher rate means a smaller a, and by (2) the slope is proportional to a. The prediction is a relation between two measured quantities with no free parameter, and it is not implied by the ordering of the rates alone. It also accounts, without any additional mechanism, for the apparent levelling off of the rate late in the delay in the condition where it is highest (R27), which the manuscript currently explains by a strategic reduction of preparation.

Saturation bounds the coherence effect, and reverses the expected interaction (R19). The maintained level is capped at θ pathwise. If a fraction $F_{D,c}$ of trials reach the threshold, conditioning (1) on that event gives

$$\nu(D,c) = G\Big[F_{D,c}\,\varphi\big(a_\theta\big) + (1 - F_{D,c})\,\mathbb{E}\big[\varphi(a) \;\big|\; |m| < \theta\big]\Big], \qquad a_\theta := \frac{g - \theta - \beta_t}{\sigma_V},$$

and as $F \to 1$ at both coherence levels only the first term survives, which does not depend on c. A larger coherence effect after a long stimulus therefore requires a substantial fraction of long-duration trials *not* to reach the threshold — and longer stimuli raise F at every coherence. The model consequently predicts a coherence effect that is *smaller* after a long stimulus, and the same parameters that would keep F low enough to avoid this also depress decision accuracy.

[2] Rigorously so for $\beta = 0$, where the ceiling $\omega/2\pi$ is exact. With a ramp there is no ceiling independent of κ, since the bracket in (1) grows with the ramp rate; for Model 2 the same region is excluded by the calibration, which returns $\kappa = 12.8$ s^{-1} against a settling requirement near one.

2.3 After the response signal

The condition effects sit before muscle activation, not after (R7–R10). Muscle activation begins when the drive reaches the gate, so premotor execution time is the passage from the starting point to the gate and carries the whole dependence on coherence and duration, through $|m|$. Motor execution time is the passage from the gate to the criterion, and after the signal the maintenance noise is off and preparation is frozen: nothing condition-dependent acts over that stretch. It should therefore be flat, and the fractionation should localise every stimulus effect in the premotor part. The one residual is the lag of the filter, still developing when activation begins early: a trial activating sooner traverses the last stretch more slowly. Evaluating that lag over the observed range of premotor times gives 2.3 ms, against a spread across the six cells of 3 ms — so the prediction is invariance, with a signed residual too small to test here.

The rate of partial activations falls (R22, R24, R25, R26). The second accumulation sweeps the signal through the gate, so a crossing is far more likely to continue to the criterion than to fall back: one crossing, and it is the response. The decline is steeper where the rate at signal onset was higher, which is the long-duration condition.

The closed forms do not apply here. The criterion is an absorbing barrier, and after the signal absorption is the rule rather than the exception; conditioning on survival destroys the Gaussianity that makes (1) explicit. This window is therefore treated qualitatively. During the delay the same issue arises but is negligible, absorption there being the anticipation rate of 1.4%.

3 From the process to the observable

Everything above concerns upcrossings of a level by a continuous process. What is measured is the output of a detection procedure applied to rectified electromyographic signals, and the two are not the same object. Three differences matter, and they are handled by applying the detection rules to the simulated signal rather than by undoing them on the data.

The count is a cluster count. `myonset` thresholds the rectified signal at seven baseline standard deviations, keeps two supra-threshold intervals separate only when the gap between them exceeds 25 ms, discards merged intervals with fewer than a minimum number of supra-threshold samples, and returns one interval per surviving cluster. Simulating the process and applying the same merge gives a ratio of clusters to upcrossings of 0.92, stable across the range of κ of interest. Any comparison between (1) and a measured rate must carry that factor, or apply the same rule to both sides.

The published counts already implement the rule. The integrated-profile step that follows moves the boundaries of each interval but does not change their number, so the exported events are already one per merged window and must not be merged a second time. Because that step can only move boundaries earlier, the event list nonetheless

contains activations separated by less than 25 ms — 27% of consecutive pairs — which is a property of the boundaries and not of the counts.

Burst durations are not usable. The refinement caps the offset at the threshold crossing while pulling the onset back before it, so a reported duration is the excursion length plus an unknown onset displacement. Durations are therefore not fitted, and the occupation identity $\Phi(-a) = \nu\bar{D}$, which would otherwise determine the distance to the gate and ω from two measurements, is used only as a bound.

What the simulation does. It applies the same rules to the simulated gated drive and counts one partial activation per surviving interval. The integrated-profile step is not simulated, since it does not affect the count.

4 The statistical re-analysis

This section carries out, for every measure and every window, the analysis the reviewer asks for.[3] Nothing is reprocessed: every value is computed from the bin means already exported in the archive, so that the published tables and the ones below rest on identical data and any difference is attributable to the contrast rather than to a re-extraction.

4.1 Method

Every value below is computed from the archive, and the extraction was checked against the archive's own exported files before anything else was done: the pooled offset-locked rate reproduces to four decimals (0.0709 per bin) and no cell deviates by more than 0.0013.

The manuscript analyses each period with a three-way repeated-measures analysis of variance, duration (2)× coherence (3)× time bin (N). Replacing the time-bin factor by its linear contrast splits that single analysis into two two-way analyses on derived scores. For each participant and each of the six cells, the N bin values $y_1, \ldots, y_N$ of that cell give

$$\text{LEVEL} := \frac{1}{N}\sum_n y_n, \qquad \text{TREND} := \sum_n w_n\, y_n, \quad w_n \propto n - \tfrac{N+1}{2},$$

with w normalised so that TREND is the ordinary least-squares slope of y against time, in units of the measure per second. Each score enters a two-way (duration × coherence) repeated-measures analysis, and TREND is additionally tested against zero. Every one of the manuscript's seven effects has an exact counterpart:

manuscript	counterpart
duration, coherence, duration × coherence	the same, on LEVEL
time	TREND against zero
duration × time	duration, on TREND
coherence × time	coherence, on TREND
duration × coherence × time	duration × coherence, on TREND

[3] Reviewer 1: "I would prefer tests on the different slopes in the EMG and EEG instead of the arbitrary 0.1s bins"; and "Ln 324: 'slope of this ramping activity' → that is not what you ANOVA tested".

Two polynomial components are tested, not one, and this is prespecified: the linear because the model predicts a monotone ramp during the delay, and the quadratic because at least one measure — mu–beta lateralisation during the stimulus — is visibly non-monotone, rising during the first second and then flattening, so that a linear contrast alone would report it as null. No higher component is tested. With two bins the quadratic does not exist and the linear component is the difference between the bins, so for the signal-locked window the decomposition reduces to the manuscript's own time effect. Where one of the two bins is structurally empty, as in the response-locked window, even that reduction is empty and the period is analysed on the informative bin alone; Section 6 explains why.

The first three effects do not involve the bin factor and are therefore *arithmetically identical* to the published ones: a main effect that ignores time is already an effect on the bin average. They are recomputed here and serve as a replication check, reported in the left-hand columns of every table below. The last four are the substantive change: each has one or two degrees of freedom instead of nineteen or thirty-eight, and each tests a monotone trend rather than any difference whatever between bins.

All of it is computed in R; Python supplies only the counts and exposures, so that the identical analysis runs on the data and on the simulated output of each model.

Two-level factors give $F(1, 21)$ and need no sphericity correction. Three-level factors give $F(2, 42)$, corrected by the Greenhouse–Geisser estimate $\varepsilon = (\operatorname{tr} \tilde{S})^2/((k-1)\operatorname{tr} \tilde{S}^2)$ with $\tilde{S} = CSC'$, C an orthonormal set of $k-1$ contrasts and S the covariance matrix of the level means across participants, as the manuscript states it did.

Windows. The bin indexing differs between the muscle and the brain export files, and the difference matters. The published values are reproduced by bins 0–19 (stimulus) and 10–29 (delay) for partial activations, and by bins 1–20 and 11–30 for both brain measures; the response-signal and response windows use two bins in every case. These were identified by searching, for each file, the window that reproduces all three published bin-independent F values simultaneously — a constraint tight enough to admit only one solution.

Replication. Eleven of the twelve analyses reproduce the published values exactly, to the last reported digit. The signal-locked window, which an earlier version of this note could not reproduce, now does: the discrepancy was an error in the exported file, since corrected, and all seven of its effects now agree ($F = 36.559$ against 36.560 for duration, 21.964 against 21.96 for time, 7.046 against 7.046 for duration × time). Two entries do not reproduce, for different reasons. The lateralised readiness potential during the stimulus differs by about three per cent on all three values in the same direction, which suggests a marginally different export; every effect there is null before and after, so nothing turns on it. The response-locked window does not reproduce because the data have changed: the corrected file gives different values, and with them a different conclusion, as Section 6 sets out.

4.2 Results

Published values on the left, linear contrast on the right. All effects are reported, including null ones.

Partial-activation rate, stimulus period.

	published	trend decomposition				
effect	F	df1	df2	F	p	η_p^2
duration	13.380	1.00	21.00	13.381	.001	0.389
coherence	5.250	1.33	28.01	5.251	.021	0.200
time, linear	18.050	1.00	21.00	30.869	<.001	0.595
duration × coherence	0.312	1.98	41.51	0.312	.731	0.015
duration × time, linear	4.645	1.00	21.00	9.532	.006	0.312
coherence × time, linear	1.718	1.72	36.20	6.145	.007	0.226
duration × coherence × time, linear	1.058	1.87	39.18	0.729	.480	0.034
time, quadratic	—	1.00	21.00	2.621	.120	0.111
duration × time, quadratic	—	1.00	21.00	3.555	.073	0.145
coherence × time, quadratic	—	1.90	39.81	0.525	.586	0.024
duration × coherence × time, quadratic	—	1.59	33.42	0.403	.625	0.019

Partial-activation rate, delay period.

	published	trend decomposition				
effect	F	df1	df2	F	p	η_p^2
duration	33.190	1.00	21.00	33.186	<.001	0.612
coherence	3.360	1.96	41.15	3.361	.045	0.138
time, linear	31.420	1.00	21.00	35.238	<.001	0.627
duration × coherence	3.474	1.92	40.42	3.474	.042	0.142
duration × time, linear	4.520	1.00	21.00	11.898	.002	0.362
coherence × time, linear	0.737	1.70	35.61	0.524	.567	0.024
duration × coherence × time, linear	0.959	1.74	36.45	0.508	.580	0.024
time, quadratic	—	1.00	21.00	18.891	<.001	0.474
duration × time, quadratic	—	1.00	21.00	0.751	.396	0.035
coherence × time, quadratic	—	1.95	40.89	1.855	.170	0.081
duration × coherence × time, quadratic	—	1.86	39.04	0.946	.391	0.043

Partial-activation rate, RS period.

	published	trend decomposition				
effect	F	df1	df2	F	p	η_p^2
duration	36.560	1.00	21.00	36.559	<.001	0.635
coherence	1.159	1.56	32.82	1.159	.315	0.052
time, linear	21.960	1.00	21.00	21.964	<.001	0.511
duration × coherence	0.041	1.91	40.04	0.041	.954	0.002
duration × time, linear	7.046	1.00	21.00	7.046	.015	0.251
coherence × time, linear	0.407	1.84	38.61	0.407	.651	0.019
duration × coherence × time, linear	0.587	1.75	36.67	0.587	.539	0.027

Partial-activation rate, response period.

	published	trend decomposition				
effect	F	df1	df2	F	p	η_p^2
duration	0.011	1.00	21.00	7.613	.012	0.266
coherence	0.724	1.90	39.98	1.845	.173	0.081
duration × coherence	0.636	1.86	39.10	0.342	.697	0.016

Mu–beta lateralisation, stimulus period.

	published	trend decomposition				
effect	F	df1	df2	F	p	η_p^2
duration	2.146	1.00	21.00	2.146	.158	0.093
coherence	2.813	1.31	27.56	2.813	.096	0.118
time, linear	6.739	1.00	21.00	11.740	.003	0.359
duration × coherence	0.755	1.62	34.00	0.755	.452	0.035
duration × time, linear	1.823	1.00	21.00	4.497	.046	0.176
coherence × time, linear	2.805	1.95	40.99	0.517	.596	0.024
duration × coherence × time, linear	0.810	1.85	38.90	0.021	.974	0.001
time, quadratic	—	1.00	21.00	9.116	.007	0.303
duration × time, quadratic	—	1.00	21.00	0.236	.632	0.011
coherence × time, quadratic	—	1.66	34.89	1.936	.165	0.084
duration × coherence × time, quadratic	—	1.99	41.79	1.302	.283	0.058

Mu–beta lateralisation, delay period.

	published	trend decomposition				
effect	F	df1	df2	F	p	η_p^2
duration	3.177	1.00	21.00	3.177	.089	0.131
coherence	1.257	1.29	27.12	1.257	.285	0.056
time, linear	5.642	1.00	21.00	7.107	.014	0.253
duration × coherence	1.586	1.69	35.43	1.586	.220	0.070
duration × time, linear	1.078	1.00	21.00	0.019	.893	0.001
coherence × time, linear	1.008	1.78	37.43	1.371	.265	0.061
duration × coherence × time, linear	1.199	1.81	37.91	2.687	.086	0.113
time, quadratic	—	1.00	21.00	3.290	.084	0.135
duration × time, quadratic	—	1.00	21.00	3.741	.067	0.151
coherence × time, quadratic	—	1.96	41.11	3.147	.054	0.130
duration × coherence × time, quadratic	—	1.29	27.01	0.565	.501	0.026

Mu–beta lateralisation, RS period.

	published	trend decomposition				
effect	F	df1	df2	F	p	η_p^2
duration	3.039	1.00	21.00	3.039	.096	0.126
coherence	0.160	1.90	39.94	0.160	.843	0.008
time, linear	0.059	1.00	21.00	0.059	.810	0.003
duration × coherence	1.216	2.00	41.96	1.216	.307	0.055
duration × time, linear	1.509	1.00	21.00	1.509	.233	0.067
coherence × time, linear	0.024	1.99	41.73	0.024	.976	0.001
duration × coherence × time, linear	1.458	1.66	34.93	1.458	.246	0.065

Mu–beta lateralisation, response period.

	published	trend decomposition				
effect	F	df1	df2	F	p	η_p^2
duration	0.006	1.00	21.00	0.006	.937	0.000
coherence	0.364	1.95	40.91	0.364	.692	0.017
time, linear	11.660	1.00	21.00	11.658	.003	0.357
duration × coherence	0.161	1.65	34.56	0.161	.811	0.008
duration × time, linear	2.129	1.00	21.00	2.129	.159	0.092
coherence × time, linear	0.252	1.98	41.51	0.252	.776	0.012
duration × coherence × time, linear	0.719	1.72	36.21	0.719	.475	0.033

Lateralised readiness potential, stimulus period.

	published	trend decomposition				
effect	F	df1	df2	F	p	η_p^2
duration	0.636	1.00	21.00	0.614	.442	0.028
coherence	1.524	1.75	36.66	1.498	.237	0.067
time, linear	1.876	1.00	21.00	0.497	.489	0.023
duration × coherence	0.391	1.87	39.26	0.387	.668	0.018
duration × time, linear	1.181	1.00	21.00	2.222	.151	0.096
coherence × time, linear	1.177	1.49	31.27	0.551	.531	0.026
duration × coherence × time, linear	0.595	1.97	41.44	0.169	.843	0.008
time, quadratic	—	1.00	21.00	0.487	.493	0.023
duration × time, quadratic	—	1.00	21.00	7.201	.014	0.255
coherence × time, quadratic	—	1.74	36.63	2.267	.124	0.097
duration × coherence × time, quadratic	—	1.93	40.61	0.161	.845	0.008

Lateralised readiness potential, delay period.

	published	trend decomposition				
effect	F	df1	df2	F	p	η_p^2
duration	0.019	1.00	21.00	0.019	.892	0.001
coherence	0.440	1.34	28.13	0.440	.569	0.021
time, linear	10.791	1.00	21.00	14.338	.001	0.406
duration × coherence	0.378	1.96	41.16	0.378	.683	0.018
duration × time, linear	0.555	1.00	21.00	0.262	.614	0.012
coherence × time, linear	0.457	1.56	32.72	0.283	.701	0.013
duration × coherence × time, linear	0.408	1.90	39.84	0.031	.964	0.001
time, quadratic	—	1.00	21.00	4.041	.057	0.161
duration × time, quadratic	—	1.00	21.00	0.574	.457	0.027
coherence × time, quadratic	—	1.93	40.62	0.682	.506	0.031
duration × coherence × time, quadratic	—	1.98	41.67	0.330	.719	0.015

Lateralised readiness potential, RS period.

	published	trend decomposition				
effect	F	df1	df2	F	p	η_p^2
duration	0.185	1.00	21.00	0.185	.671	0.009
coherence	0.252	1.45	30.43	0.252	.707	0.012
time, linear	14.076	1.00	21.00	14.076	.001	0.401
duration × coherence	0.267	1.96	41.08	0.267	.762	0.013
duration × time, linear	1.238	1.00	21.00	1.238	.278	0.056
coherence × time, linear	1.058	1.59	33.39	1.058	.344	0.048
duration × coherence × time, linear	0.108	1.94	40.83	0.108	.893	0.005

Lateralised readiness potential, response period.

	published	trend decomposition				
effect	F	df1	df2	F	p	η_p^2
duration	0.400	1.00	21.00	0.400	.534	0.019
coherence	0.310	1.55	32.45	0.309	.680	0.014
time, linear	54.331	1.00	21.00	54.331	<.001	0.721
duration × coherence	0.666	1.97	41.30	0.666	.517	0.031
duration × time, linear	4.129	1.00	21.00	4.129	.055	0.164
coherence × time, linear	0.246	1.60	33.70	0.246	.734	0.012
duration × coherence × time, linear	0.589	1.62	34.05	0.589	.526	0.027

4.3 What changes

Partial activations: everything survives, including the decisive null (R13–R15, R18, R20, R21). During the stimulus the trend is positive, steeper at short duration, and steeper at high coherence — the last being R15, the designated diagnostic marker of evidence accumulation, which passes at $p = .007$ against $p = .005$ for the omnibus while costing two degrees of freedom instead of thirty-eight. During the delay the trend is positive and steeper at long duration ($F(1, 21) = 11.90$, $p = .002$), and the coherence × time interaction remains null ($F(1.70, 35.61) = 0.52$, $p = .567$, against $p = .879$). That null

is what rejects the threshold-only account, and it is not an artefact of a diffuse omnibus test: the rate ramps through the delay without a coherence-dependent slope, under either analysis.

The quadratic component. It is carried because mu–beta lateralisation during the stimulus is not monotone and a linear contrast alone would report it as null. For partial activations it finds a curvature during the delay, $F(1, 21) = 18.89$, $p < .001$, $\eta_p^2 = 0.474$, and nothing during the stimulus ($F = 2.62$, $p = .120$). It is a precaution against non-monotone profiles, applied identically to the three measures, and not a test of the model.

Mu–beta lateralisation: one claim changes, and one effect appears. The coherence × time interaction, $p < .001$ in the omnibus, has *no linear component over the full stimulus window* ($F = 0.52$, $p = .596$). Resolving the window into two halves shows why:

	trend	trend × coherence	trend × duration
first ten bins	$F = 11.19$, $p = .003$	$F = 5.76$, $p = .018$	$F = 5.67$, $p = .027$
last ten bins	$F = 2.22$, $p = .151$	$F = 0.37$, $p = .658$	$F = 5.34$, $p = .031$
full window	$F = 11.74$, $p = .003$	$F = 0.52$, $p = .596$	$F = 4.50$, $p = .046$

Cell trends over the first half are -0.029, -0.079, -0.083 at short duration and -0.011, -0.032, -0.078 at long: monotone in coherence in both, and since the measure is negative-going with preparation, a steeper buildup at higher coherence. Over the second half the ordering is gone and the trends are near zero. The effect is not reversed, it is *extinguished* — which is what the saturation argument of Section 2 requires, since trajectories that have reached the internal threshold are pinned at a common level and their slope can no longer depend on coherence. The manuscript's sentence should be stated in two phases rather than one, and it then reports evidence for the account rather than a claim its own test does not support. Choosing the cut-off after inspecting the contrast would be a forking path, so it should be fixed a priori from the model's commitment-time distribution.

The duration × time interaction moves the other way: $p = .046$ under the contrast against $p = .136$ in the omnibus, and it holds in both halves separately. A monotone effect that thirty-eight degrees of freedom had diluted becomes visible at one.

Lateralised readiness potential: unchanged. Flat during the stimulus, ramping during the delay ($p = .004$), after the response signal and before the response, with no dependence on coherence or duration anywhere. The contrast reproduces this picture exactly and adds nothing, which is itself worth stating: the change of analysis does not manufacture effects.

5 Two analyses the manuscript does not contain

The re-analysis above replaces the manuscript's own tests. The two analyses below add to them: an estimate of the rate that uses no bins at all, and a test of a relation between two measured quantities that no analysis of variance expresses.

5.1 The rate without bins

The reviewer's objection to the bin factor is well taken, and the quantities in (2) do not require bins at all. The onsets of partial activations are a point process on the delay window with intensity $\nu(t)$. Writing $\ln \nu(t) = c_0 + c_1 t + c_2 t^2$, the log likelihood of an inhomogeneous Poisson process observed on $[T_0, T_1]$ is

$$\ell(c) = \sum_i \ln \nu(t_i) - \int_{T_0}^{T_1} \nu(t)\, \mathrm{d}t, \tag{3}$$

a function of the exact event times, from which c_1 and c_2 are read off directly. Three remarks.

Three estimands, not one. They are not interchangeable, and this note uses all three. A *linear contrast* on the raw rate estimates $\mathrm{d}\nu/\mathrm{d}t$, an additive slope, and is what Section 4 reports. A *Poisson regression with a log link*, binned or not, estimates $\mathrm{d}\ln\nu/\mathrm{d}t$, a multiplicative slope, and is what (2) predicts. The *point-process likelihood* (3) is the unbinned form of the second, not of the first.

The two scales can disagree, and they do. During the stimulus period the coherence effect on the slope is significant on the raw scale ($p = .007$) and absent on the log scale ($p = .25$), because the log slope divides by the level and the high-coherence cell has a much higher rate, so the same additive rise is a smaller relative rise. That is not a power problem: the log slopes are non-monotone in coherence at short duration (0.550, 0.782, 0.703) where the raw contrasts are monotone. Neither scale is privileged during the stimulus, because the model predicts no functional form there. During the delay it predicts a log-linear one, which is why the point-process analysis is restricted to that window.

Bins are harmless if they are an offset rather than a factor. A Poisson regression of the count in a bin of width w on the bin centre, with $\log(w \times n_{\text{trials}})$ as offset, is the discretised form of the same estimating equation, because $\mathbb{E}[\text{count}] = w\nu(t)\big[1 + \frac{w^2}{24}\big((\ln\nu)'' + ((\ln\nu)')^2\big) + O(w^4)\big]$; at $w = 0.1$ s and the values estimated here that is a relative error of 2.5×10^{-4} on the rate and none on the slope. The bin width is therefore not an arbitrary analytic choice in the sense the reviewer objects to. What was arbitrary was entering the bin as a nineteen-level factor.

The process is not Poisson, so the likelihood is a quasi-likelihood. Upcrossings cluster: an excursion that dips briefly below the gate and returns produces two events a few milliseconds apart, which is also why the detection procedure merges them. Counts are consequently overdispersed and the Poisson information understates the variance. The Poisson likelihood is therefore best read as a quasi-likelihood, or an estimating equation for the first-order intensity, rather than as the likelihood of the process. Standard errors must come from between-participant variation. A further caveat belongs here: what the detector delivers is the intensity of clusters, which is the upcrossing intensity times a retention factor slightly below one, and that factor depends on the distance to the gate and so drifts across the window. The measured log-rate slope is thus the sum of the quantity of interest and the derivative of log of that factor, which is small but not identically zero.

Clustering also enters the intensity. What the detector delivers is the intensity of clusters, which is the upcrossing intensity times a factor slightly below one. That factor

depends on the distance to the gate and so drifts across the window, contributing a small bias to the estimated slope.

Results. Two estimators are computed, in R, and they agree. The two-stage version fits the intensity separately for each participant and cell and enters the coefficients into the same repeated-measures analysis as any other score, so that the second level supplies the standard errors and the clustering does not bias inference. The one-stage version fits a single log-linear model to all participants with cluster-robust standard errors.

	two-stage	one-stage, robust
log-rate slope	$+0.929$, $F(1, 21) = 102.95$, $p < .001$	$+0.841$, $z = 11.23$, $p < .001$
slope × duration	$F(1, 21) = 4.78$, $p = .040$	-0.179, $z = -3.17$, $p = .002$
slope × coherence	$F(1.38, 29.04) = 4.20$, $p = .038$	-0.263, $z = -5.02$, $p < .001$
slope × both	$F(1.87, 39.23) = 0.39$, $p = .663$	—

The slope is positive, shallower after a long stimulus and shallower at high coherence — both of which follow from (2), since either raises the maintained level and so reduces the distance to the gate. The coefficients differ between the two estimators by about a tenth, which is expected: the two-stage estimator weights participants equally and the one-stage weights them by precision. A difference in sign would not have been expected.

5.2 A shape prediction

The delay window supports one test that no bin-wise analysis expresses, because it is a statement about the relation between a cell's rate and the shape of its profile. By (2) the log-rate slope is $b\,a(t)$, so a cell whose maintained level is closer to the gate must ramp more slowly. Within participants, the regression of slope on log rate across the six cells returns -0.386 ± 0.070, $t(21) = -5.5$: the prediction holds, and it does not follow from the ordering of the rates alone.

6 What changed with the corrected data

Two of the exported files contained errors and have been replaced. This section records every result that moved, so that nothing is silently updated. The stimulus and offset-locked windows are unaffected: their files were correct, and every number in this note that depends on them stands.

6.1 The signal-locked window now reproduces

The earlier file did not reproduce the published time effects, and this note set the period aside for that reason. With the corrected file all seven effects agree with the manuscript to the last digit:

effect	published	recomputed
duration	36.56	36.559
coherence	1.159	1.159
time	21.96	21.964
duration × coherence	0.041	0.041
duration × time	7.046	7.046
coherence × time	0.407	0.407
duration × coherence × time	0.587	0.587

Nothing in the manuscript changes here. What changes is that the period can now be used, and that the analysis machinery is validated on it.

6.2 The response-locked window changes conclusion

The corrected file gives, in the bin from −0.2 to −0.1 s before the button press, rates of 0.021, 0.020, 0.023 after a short stimulus and 0.026, 0.027, 0.033 after a long one, and *exactly zero* in the bin from −0.1 to 0 s. The zero is mechanically correct: the last hundred milliseconds lie entirely inside the response burst, and a partial activation is by definition an activation preceding that burst. The earlier file gave rates near 0.10 in both bins, which cannot be right.

effect	published	corrected data
duration	$F = 0.011$, $p = .919$	$\mathbf{F = 7.61}$, $\mathbf{p = .012}$
coherence	$F = 0.724$, $p = .491$	$F = 1.85$, $p = .173$
time	$F = 38.31$, $p < .001$	$F = 71.75$, $p < .001$
duration × time	$F = 5.013$, $p = .036$	$F = 7.61$, $p = .012$

The main effect of duration passes from null to significant. The partial-activation rate in the two hundred milliseconds before the response is higher after a long stimulus, where it appeared not to be. Three consequences.

The manuscript's Table 2 and Table S2 must be updated for this period, and the sentence attributing the response-period pattern to the rate at signal onset should be revisited.

The oddity noted in the provenance document — that the duration main effect is significant in the signal-locked window but not in the response-locked window, which made the explanation offered for R25 hard to sustain for the response-locked window — *disappears*. The two periods now agree, and the account given for one carries over to the other. This is a gain, not a loss.

And the period should no longer be analysed as a temporal profile at all. With one bin identically zero the two scores are the same number up to a constant — with two bins of 0.1 s, TREND = −20 LEVEL — and a repeated-measures analysis is invariant under rescaling of the dependent variable, so every time-related effect returns the F of the corresponding main effect. Reporting seven effects would be reporting three of them twice. Worse, "duration × time" would be read as a difference in the *rate of decline*, when

it says only that the rate in the single informative bin differs by duration, which is what the main effect already says.

The period is therefore analysed on the informative bin alone, with three effects:

effect	F	p	η_p^2
duration	7.61	.012	0.266
coherence	1.85	.173	0.081
duration × coherence	0.34	.697	0.016

The emptiness of the second bin is structural rather than empirical: a partial activation precedes the response burst by definition, and the burst begins about 160 ms before the button press, so the last hundred milliseconds cannot contain one. No parameter setting, and no analysis, could have produced anything else there.

What this costs, and what it does not. The manuscript reads this period as the decline of the partial-activation rate once the second accumulation begins, which is R18 and R24. That decline is not in question — but it is established by the *response-signal* period, where both bins are informative (0.13 then 0.10, $F = 21.96$) and where duration × time is a genuine difference in slope ($F = 7.05$, $p = .015$). The response-locked window added nothing to it, using a window that could not test it. What the period does contribute is the duration effect on the rate just before the press, which is a real result and which reconciles R26 with R22.

The alternative, not taken. Re-exporting the window as three bins, −0.3 to 0 s, would restore two informative points and with them a meaning for the slope. We do not recommend it: the signal-locked window already carries the argument on a correct window, and lengthening this one to recover a slope that exists elsewhere is redundant.

Consequence for the manuscript (R26). The "Response" row of Tables 2 and S2 changes shape as well as value: three effects instead of seven, and the Methods must say that this period is analysed on a single bin. That is visible to a reviewer, so it is better announced than discovered — the more so since it follows from a data correction that will be reported in any case.

Everything else in this note rests on the onset-locked and offset-locked windows or on the event files, none of which changed.

7 Behaviour does not discriminate between the two models

The manuscript states that behaviour provided limited leverage for distinguishing the two candidate accounts, and moves to the physiological measures on that basis.[4] The claim can now be made quantitative rather than asserted, and it comes out stronger than the verbal version.

[4] Reviewer 2: "the authors should implement and simulate their extended model and show that the main qualitative predictions hold across a range of parameters".

Each model was calibrated *independently and on behaviour alone*: accuracy, mean response time, the interquartile range of response time, and the proportion of anticipatory responses, in each of the six cells. No electromyographic or electroencephalographic quantity entered the criterion. The two fits were then evaluated on the offset-locked partial-activation profile, which neither had seen.

criterion block	Model 1	Model 2
accuracy	8.95	8.95
mean response time	**0.73**	0.87
response-time interquartile range	**1.72**	2.16
anticipatory responses	5.63	**2.16**
total	17.03	14.13

On accuracy the two are identical to the second decimal, because both share the same over-prediction at twelve per cent coherence and two seconds. On response time *Model 1 fits slightly better*: it returns 0.558, 0.557, 0.500, 0.512, 0.500, 0.503 across the six cells against observed 0.561, 0.550, 0.521, 0.509, 0.493, 0.484, and reproduces both the coherence effect and the duration effect — the two findings that motivate the entire framework.

The two effects that motivate the paper therefore do not discriminate between the two accounts. What separation there is sits in the anticipation rate, where Model 2 is better by a factor of two and a half — and that is a weak reed. The two models differ there by less than they differ from the data; each fitted on its own gives a mean rate close to the observed 1.39% (0.93% and 1.20%), so the difference is in the pattern across cells rather than in the level; a shift in the commitment threshold absorbs the effect; and the manuscript excludes anticipatory responses from analysis rather than treating them as a dependent variable.

The same two fits, evaluated on the offset-locked log-rate slope they were not fitted to:

	slope by cell (s^{-1})						error
Model 1	+0.05	+0.01	+0.04	+0.02	+0.06	−0.04	67.3
Model 2	+0.55	+0.67	+0.59	+0.64	+0.58	+0.63	**7.4**
observed	+1.03	+1.00	+0.92	+0.88	+0.87	+0.66	

A factor of nine. Model 1 fitted to behaviour alone predicts a offset-locked slope indistinguishable from zero; Model 2 predicts one of the right order, though still too shallow. This is the argument the manuscript wants, in its strongest form: it is not that behaviour is uninformative, it is that behaviour is fitted *equally well* by both accounts while the muscle separates them by an order of magnitude.

8 Simulations and calibration

The reviewer asked for a quantitative version of the model rather than the qualitative cartoons of Figure 1.[5] This section supplies it: a single parameter vector, fitted once, evaluated against every quantity the electromyography delivers.

8.1 What is fixed and what is free

Only σ_x is fixed, at 0.100, and it fixes the amplitude unit: the model is invariant under a common rescaling of x, y, θ, g, r and SD_ε, so one amplitude has to be declared. Nothing else is inherited. In particular the response criterion r and the residual motor latency L_{motor}, which previous work fixed from immediate-report fits, are free here and constant across conditions; Section 8.3 reports what the data make of them, and it is not what was assumed.

Three combinations are weakly conditioned rather than unidentified. Only $\lambda\kappa$ enters the crossing rate, and λ is separated from κ only through the filter lag during accumulation and through a post-signal transient worth about two milliseconds; β and the two encoding latencies enter the ramp only through $\alpha := \beta/\Delta$ with $\Delta := 3 + L_{\mathrm{RS}} - L_{\mathrm{motion}}$; and L_{motor} and r/μ_{RS} enter the mean of motor execution time only through their sum. All three are fitted, and the note reports the combinations rather than pretending the individual values are well determined.

8.2 Procedure

Stage A, closed form. The identities of Sections 2 and 3 determine a starting point with no search at all. The crossing rate and the distance give ω; the mean of motor execution time gives r/μ_{RS}; the mean of premotor execution time gives $(g - |m| - \beta)/\mu_{\mathrm{RS}}$; the ratio of the two, divided by the distance, gives r/σ_V; and the delay log-rate slope gives α. This is worth doing for its own sake: it shows which measurement carries which parameter, and it starts the search at an objective of 188 rather than at an arbitrary point.

Stage B, search. Nelder–Mead on all thirteen free parameters in logarithmic coordinates, with common random numbers so that the objective is deterministic, and repeated restarts with progressively larger initial simplices to check that the optimum is not an artefact of a collapsed simplex.

The objective. Nine blocks of six cells: accuracy; the mean and the interquartile range of premotor execution time; the mean and the interquartile range of motor execution time; the offset-locked partial-activation rate; the offset-locked log-rate slope; the onset-locked rate; and the premature-response rate. Each block is divided by the between-participant standard error of the corresponding target, so one unit of error is one standard error, and blocks measured in seconds and blocks measured in counts are commensurable.

Dispersion is measured by the interquartile range rather than the standard deviation. The empirical distribution of premotor execution time has a long right tail — 2.7% of trials exceed one second and the ninety-ninth percentile is 1.50 s — which inflates the standard deviation to 0.264 s against a trimmed value of 0.113 s. The model has no

[5] Reviewer 2: "Overall, it seems that the hypotheses could be evaluated more quantitatively, based on a simulated model rather than the qualitative representations in Fig 1."

mechanism for that tail, because it has no across-trial variability in the rate of response-signal accumulation, and scoring it against the untrimmed standard deviation forces every other parameter to manufacture variability elsewhere. Using the interquartile range moved the objective from 71 to 45 without changing a single parameter. The tail is a real limitation and is recorded as such rather than absorbed.

The detection rules are applied to the simulation, not undone on the data, as Section 3 sets out; burst durations are not fitted, for the reason given there.

8.3 Result

parameter	fitted	previously assumed
k	1.109	1.183
λ	25.5 s^{-1}	21.471
θ	0.0707	—
g	0.1521	—
κ	12.8 s^{-1}	—
SD_ε	0.0430	—
α	0.0135 s^{-1} ($\beta \approx 0.041$)	—
μ_{RS}	0.574	—
σ_q	0.0804	0.100
r	**0.0694**	**0.022**
L_{motor}	**0.019** s	**0.093** s
L_{motion}	0.190 s	0.200
L_{RS}	0.215 s	0.200

The timescale agrees with three independent routes. $\lambda\kappa = 325\ s^{-2}$, that is $\omega = 18.0\ s^{-1}$. The analytic lower bound from the rate alone was 240; the occupation-time recovery, corrected for the onset displacement, gave 11 to 15 for κ at $\lambda \approx 21.5$. The numerical optimum was not constrained towards either, and lands with them.

Two inherited constants do not transfer. The response criterion comes out three times its assumed value and the residual motor latency five times smaller. This is not a preference of the fit; it is forced. The ratio r/σ_V must be near two to keep premature responses at the observed 1.4% given the observed partial-activation rate, and that ratio equals $a\,(\overline{\mathrm{MET}} - L_{motor})/(\overline{\mathrm{PMET}} - L_{RS} - 1/\lambda)$. A residual motor latency of 93 ms leaves no room. A residual latency of 19 ms is in turn shorter than the electromechanical delay usually reported, and the onset displacement introduced by the integrated-profile step inflates the measured motor execution time in the same direction. This should be reported as a tension, not as an estimate.

Table S1, cell by cell. The quantities the manuscript reports in Table S1, observed against the calibrated Model 2:

	accuracy		response time		premotor time		motor time	
cell	obs	model	obs	model	obs	model	obs	model
0.3/c00	.494	.480	.561	.545	.399	.393	.162	.152
0.3/c12	.708	.758	.550	.531	.390	.372	.160	.159
0.3/c50	.948	.997	.521	.484	.361	.319	.159	.165
2.0/c00	.507	.511	.509	.488	.349	.326	.160	.162
2.0/c12	.788	.919	.493	.480	.332	.319	.161	.161
2.0/c50	.946	1.000	.484	.482	.323	.318	.161	.164
duration	−.031	−.065	+.049	+.037	+.049	+.040	−.001	−.004
coherence	+.447	+.503	−.033	−.034	−.032	−.041	−.001	+.007
interaction	+.014	+.028	−.015	−.055	−.011	−.066	−.003	+.011

All ten results of Table S1 come out with the right sign. The three main effects on response time and premotor time are close in magnitude; motor time is flat in the model as in the data, the largest simulated effect being four milliseconds. Two things are off. The interactions on response and premotor time (R6, R9) are right in sign but four to six times too large, which is the cap on the maintained level biting harder in the model than in the data. And accuracy at two seconds and twelve per cent comes out at 0.92 against 0.79, which is discussed below.

Where the fit fails. Three residuals are structural rather than noise. Accuracy at two seconds and twelve per cent coherence comes out at 0.90 against 0.79 observed. The offset-locked log-rate slope is too shallow throughout, 0.54 to 0.68 against 0.66 to 1.03. And the offset-locked rate has the wrong coherence structure at short duration: the model gives 0.023, 0.029, 0.075 across the three coherence levels where the data give 0.054, 0.055, 0.058, and it is flat at two seconds where the data rise. This last one is the saturation argument of Section 2 made numerical, and it is discussed in Section 9.

8.4 Model 1, recalibrated in its own right

Model 1 was fitted independently, with its own best values for every parameter, rather than evaluated at Model 2's. That is the fair comparison: sharing a parameter vector would handicap it on overall scale and confound the shape of the delay profile with its level.

	Model 1	Model 2
total objective	160.3	**47.3**
delay log-rate slope, error	81.7	**6.5**
fitted slopes across the six cells	−0.17 to +0.12	+0.54 to +0.71
observed	+0.66 to +1.03	

Model 1's offset-locked slopes are indistinguishable from zero and sometimes negative, against a ramp that is one of the most robust effects in the data. Its fitted κ is 14.5 s^{-1}: the search did not drift into the slow-reversion regime where the non-stationary transient could manufacture a ramp, because doing so would destroy the rate. The discrimination between the two variants therefore does not depend on the parameterisation.

8.5 Robustness

Each free parameter was multiplied by one half and by two in turn, everything else held at the calibrated value, and every directional prediction re-evaluated; a leak common to both accumulators was added as a further variant at the three rates the manuscript already uses.[6]

	R7	R8	R10	R16	R17	R18	R19	R20
Model 2	30/30	30/30	21/30	28/30	28/30	**29/30**	**3/30**	5/30
Model 1	30/30	30/30	24/30	28/30	28/30	**1/30**	1/30	1/30

Three readings. The offset-locked ramp separates the two variants completely and does so at every setting, including all three leak rates. R19 fails in Model 2 almost everywhere, and the three settings where it holds are those with a raised threshold or a reduced fluctuation, which is what the saturation argument of Section 2 predicts: R15 needs a substantial fraction of long-duration trials to stay below the threshold. And the low count for R20 in Model 2 is a power artefact of the sweep rather than a failure — the sweep uses 150 trials per cell, at which a slope difference is not resolved; at 700 trials the effect has the observed sign at the calibrated point.

9 What does not follow

R19. By the saturation argument of Section 2 the model produces a coherence effect on offset-locked rate that is *smaller* after a long stimulus, whenever long stimuli push more trials to the threshold — which they do at every parameter setting for which decision accuracy is in the observed range. The observed interaction has the opposite sign. The manuscript should report this as a failure rather than as support, and the Discussion's threshold-reaching account of it should be withdrawn, since it also contradicts the account given for R6 and R9.

R26, which may not be assessable. The response-period window ends at the button press, and its content therefore depends on how the response burst is segmented. If a single activation is sometimes split in two by the detector during the rise to the response, the earlier fragment is counted as a partial activation inside the window, and the response-period effects are partly a segmentation artefact. Whether that happens is a question about the detection procedure, not about the model.

10 Point-by-point response to the reviews

Each comment is quoted, followed by what this note supplies and what remains to be decided. Comments the modelling does not bear on are marked *outside this note*, with a suggested reply.

[6] Reviewer 2: "the authors should show that the predictions hold over a range of parameters"; and "I wonder how the model would behave with a leak in the accumulation process".

10.1 Reviewer 1

C1.1 Slopes rather than bins. *"Although there is a nice cartoon figure for the hypotheses, the authors use a repeated measures ANOVA for their tests. I would prefer tests on the different slopes in the EMG and EEG instead of the arbitrary 0.1s bins."* And *"Ln 324: 'slope of this ramping activity' → that is not what you ANOVA tested."*

Answered: Sections 4 and 5. The reply that emerged from the analysis is not the one an earlier draft proposed, and it is more defensible. The comment combines two issues, and they have different answers.

The second half is simply right and should be conceded: an omnibus F on a nineteen-degree-of-freedom factor establishes that bins differ, not that the rate ramps with a coherence-dependent slope. Either the language changes or the test does.

For the first half, the choice of test should follow the data type rather than be uniform. For the brain measures the signal exists at every sample and the bin is a *summary*; for partial activations the bin is what *creates* the dependent variable, since a rate does not exist without aggregation. The reviewer's objection to arbitrary bins therefore bites hardest exactly where it can be removed, and least where it cannot. We recommend: keep the categorical analyses as primary everywhere, so that the muscle and brain analyses remain aligned; add linear contrasts of the existing time factor as a supplement, which are weighted sums of bin means already computed and require no reprocessing of anything; and, for the offset-locked partial activations only, add the point-process intensity estimated from exact event times, which removes the bin entirely. The delay is singled out on principle rather than convenience: Section 2 implies that the onset-locked rate is the commitment-time distribution convolved with post-commitment dynamics, so a parametric intensity is misspecified there by the theory's own account, and empirically the onset-locked coherence effect does not survive a log-linear intensity though it survives the linear contrast.

All six reported time effects on partial activations reproduce under the linear contrast, including the offset-locked coherence × time null that carries the argument against the threshold-only account. An earlier draft of this note claimed that the offset-locked coherence × time interaction becomes significant on the log scale; that was an error and is withdrawn.

C1.2 Clearer statement of the hypotheses and of what each measure does. *"I would like a much clearer description of the hypotheses and how the different measures help unraveling. Now the EEG results are a bit of a hotch-potch."*

Partly answered: Sections 8 and 4. The sweep separates the effects that discriminate from those that do not: R7, R8, R10, R16 and R13 hold in both variants at essentially every setting and therefore carry information about the architecture rather than about temporal preparation, while R11, R14, R18 and R21 separate the two variants completely. *Suggested addition, outside this note*: reorganise the Results so that each measure is introduced with the hypothesis it adjudicates, and state explicitly that the shared effects establish the architecture rather than the preparation ramp.

C1.3 Muscle-activation onset procedure. *"Ln 119: please be more explicit about the EMG onset procedure."* Also Reviewer 2's minor point on the same line, and *"p. 10,*

ll. 157-159: PMA should be more precisely defined. Specifically, how is it decided that a muscle activation detected by EMG does not 'culminate in an overt response'?"

Now answerable in full, and it should be. The procedure is `myonset` (`detector_var`) with a threshold of seven baseline standard deviations on the rectified signal, merging of supra-threshold intervals separated by at most 25 ms, a minimum of three supra-threshold samples per merged interval, followed by the integrated-profile refinement of the boundaries. Four sentences suffice, and they are necessary: these rules define the dependent variable of half the paper, and the manuscript currently cites only "the three-step semi-automatic procedure".

Two consequences worth stating. First, because the refinement can only move boundaries earlier, the published event list contains activations separated by less than 25 ms — 27% of consecutive pairs — although the rule was applied; the counts are nonetheless correct, since the refinement does not change the number of intervals. Second, an activation is judged not to culminate in a response by being an interval other than the last one before the button press, which makes the response-period analysis sensitive to segmentation: 18.5% of trials contain an activation ending within 25 ms of the response burst, and 9.3% would be absorbed into it under a strict application of the merge rule. The response-locked results should be reported with that caveat.

C1.4 Coherence range used by others. *"Ln 242: make explicit what range of coherences others have used." Outside this note.* A sentence in the Discussion with two or three references.

C1.5 Error bars and individual data. *"Fig 3: SEM is not a good measure on data constrained between 0 and 1... In modern research it is custom to also show individual participant datapoints." Outside this note.* Per-participant points with connecting lines, and for accuracy a logit-scale or bootstrap interval rather than a symmetric standard error.

C1.6 Reminder of MET and PMET. *"Ln 308: please refer back to fig 2 to reiterate what MET and PMET are." Outside this note*; a cross-reference.

C1.7 Different delays, to show build-up. *"Ln 485... If you really want to hammer in this point, you should show build-up for different delays (even within the same trial, where the RS can come at for example 1.5 or 3.0 sec)."*

Outside this note, but the experiment exists. The companion experiments vary the delay at fixed stimulus duration, with the delay length known in advance or not. Two findings speak directly to the request: with foreknowledge the delay profiles superpose on the fraction of the delay elapsed rather than on time since offset, so the preparation slope is scaled to the expected delay; and without foreknowledge the build-up disappears. If those data can be cited they answer the request without a new experiment. Note also that this makes α a function of the delay length, invisible here because the delay is fixed, and it should be stated as a scope condition.

C1.8 A quantitative model. *"Ln 505-506: here you hint at a quantitative model, which I believe would strengthen your data better. Now you use ANOVAs, and thus significances to rule out hypotheses that are only expressed in words or cartoons, but are never quantified based on the data."*

Answered: Section 8. A single parameter vector, fitted once against nine families of measurements, with the detection rules of the recording applied to the simulated signal; and the competing variant refitted independently rather than evaluated at the winner's parameters.

C1.9 Statistics in the text. *"In J Neurosci it is customary to present statistics in the results text instead of everything in a table." Outside this note*; a formatting change.

10.2 Reviewer 2

C2.1 Simulate the model; show the predictions hold across parameters. *"the predictions in Fig. 1 are qualitative rather than quantitative and seem to be based on simple drawings of hypothetical scenarios rather than simulations of actual models. This does not hold up to the standards of the field of decision neuroscience. The authors should implement and simulate their extended model and show that the main qualitative predictions hold across a range of parameters."*

Answered: Sections 8. Every free parameter varied by factors of one half and two, thirty settings including three leak rates, every directional effect re-evaluated. R7 and R8 hold at every setting; R16, R17 and R18 at 28 to 29 of 30; R19 at 3 of 30, for the reason given in Section 9.

C2.2 The specific ordering in Fig. 1. *"the slopes of the accumulator dynamics for the two motion coherence conditions are unknown, only their ordering can be safely assumed. Thus, it is not clear whether a long low-coherence stimulus yields a larger cumulative signal than a short high-coherence stimulus, as indicated in Fig. 1."*

Answered: Section 8. At the calibrated parameters the mean maintained level is 0.0701 after a long twelve-per-cent stimulus and 0.0690 after a short fifty-per-cent one, so the ordering drawn in Figure 1 holds — but barely, and for an instructive reason: 99% of long twelve-per-cent trials and 93% of short fifty-per-cent trials reach the internal threshold, so both are close to being pinned at $\theta = 0.0707$ and the ordering is nearly an equality. The right reply is therefore not that the ordering is assumed but that it is a near-tie produced by saturation, and that saturation is itself testable.

C2.3 Leak. *"This is even more true once a non-zero leak is introduced in the integrator, as in generalized diffusion models."*

Answered: Section 8. A leak common to both accumulators at 0.25, 0.50 and 1.00 s^{-1} changes no directional prediction in either variant. It is worth adding, as the reviewer implicitly does, that the architecture already contains a first-order filter between accumulator and muscle, which is a leak on the transmitted signal; what the reviewer asks for is a leak on the accumulator itself, and that is what was added.

C2.4 What is accumulated in the second phase. *"the response signal is unambiguous and presented at fixation, leaving no sensory evidence to be accumulated before it can be detected even in the presence of strong internal noise. This aspect should be unpacked."*

Answered: Section 1. The reviewer is right, and the model does not need the second phase to be perceptual accumulation. What it needs is a drive with independent increments, common to all conditions, released once the signal is detected. Two properties are used, and only two: the drive being condition-independent puts all condition effects on premotor execution time into the starting point, and the increments being independent make motor execution time invariant (Section 2). We suggest renaming the stage — a motor drive or a rise to threshold rather than a second accumulation of evidence — which costs nothing and removes the objection.

C2.5 Scholarship: motor and premotor circuits. *"studies from the lab of Karel Svoboda have consistently identified a 'ramping mode' of non-selective drive of mouse (pre)motor cortex... both the Svoboda work in mice as well as work in monkey indicate that effector-selective motor preparatory activity and activity driving motor execution 'live' in different subspaces."*

Outside this note, but the model makes the link concrete. The architecture has exactly the two components described: a selective maintained level, which carries the choice and does not by itself trigger movement, and a non-selective drive released at the response signal. Gating inhibition is the model's version of the separation of subspaces. A paragraph making the correspondence explicit also strengthens R2.4, since it supplies the mechanistic reading of the second stage.

C2.6 Writing: abbreviations and placement of the reasoning. *"the text contains a large number of abbreviations, many of which are non-standard... much of the key reasoning underlying the hypotheses is folded into the EMG part of the Methods section." Outside this note.* We suggest moving the derivation of the predictions into the Results, where the simulated figures now provide a natural anchor.

C2.7 Broader evidence on selective motor preparation. *"p. 21, l. 418... I suggest to extend it by also incorporating the accumulation of sensory evidence in other formats and to animal work." Outside this note*; additional citations.

10.3 Editorial note

"in framing the dynamic regulation of action during working memory and decision making, the authors may also wish to consider other work by van Ede." Outside this note; citation.

10.4 Four things we recommend volunteering

None is asked for, and all are better raised by the authors than discovered by a reviewer running the code, which is now possible since the data and analysis code are public.

R15. The duration × coherence interaction on offset-locked rate comes out with the wrong sign in almost every parameter setting, and the saturation argument of Section 2 shows why. At the calibrated point the model gives a coherence effect of 0.052 per bin after a short stimulus and 0.009 after a long one; the data give 0.004 and 0.015. Escaping this requires a substantial fraction of long trials not to reach the threshold, which the accuracy data forbid. We suggest reporting R19 as a limitation with the mechanism stated.

The two branches invoked for R6, R9 and R19. The Results explain R6 and R9 by coherence differences in the starting point being *smaller* after a long stimulus; the Discussion explains R19 by long stimuli *amplifying* coherence differences in maintained activity. These concern the same latent quantity and predict opposite interactions. The manuscript's own Methods say the direction is not predicted in advance. We suggest resolving this in one direction and presenting R6 and R9 as evidence against the temporal-preparation hypothesis rather than as evidence for the combined one.

Two inherited constants do not transfer. The response criterion and the residual motor latency taken from immediate-report fits are incompatible with the data by roughly a factor of three and a factor of five (Section 8.3). Since the manuscript states that these values were inherited, the revision has to either free them or explain the discrepancy. Freeing them costs little: continuity with immediate-report models does not depend on numerical identity of every constant.

A significant effect that is never mentioned. The main effect of duration on onset-locked partial-activation rate, R11, is significant with a large effect size and appears in Tables 2 and S2, but no sentence of the Results or Discussion refers to it. The theory explains it with no additional assumption — it is the average counterpart of the duration × time interaction the manuscript does discuss — so adding a sentence costs nothing and removes the obvious question of why a reported effect goes unremarked.

The mu–beta buildup-rate sentence. As written, the claim that mu–beta lateralisation showed a coherence-dependent buildup rate has no linear component over the full stimulus window. It has one over the first second, after which the coherence dependence vanishes — which is saturation, and therefore evidence for the account rather than against it. Restating the sentence in two phases is stronger than leaving it as is, and it removes an easy target.

11 Figures

A Tables S1 and S2, with the observations identified

The two tables below are Tables S1 and S2 of the manuscript, with one column added at the left. A label Rx marks a result this note discusses; a blank means the result is a null that the manuscript never mentions, and there is nothing to say about it. Values are the

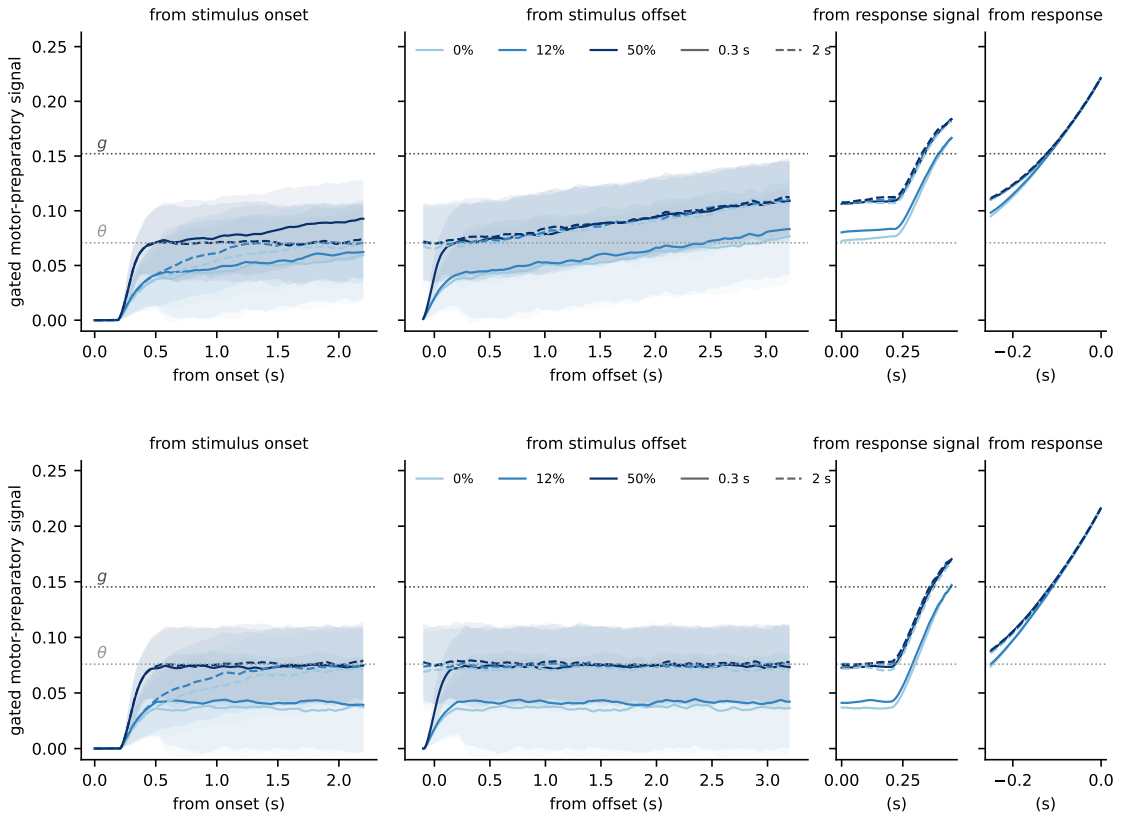


Figure 1: The manuscript's Figure 1, simulated. Mean gated motor-preparatory signal at the calibrated parameters, time-locked to the four trial events, with a one standard deviation band; each trial is censored at its own response. Colour is coherence, line style is stimulus duration; the dotted lines are the internal threshold θ and the gate g. Top: Model 2. Bottom: Model 1, refitted in its own right. The difference is entirely in the second panel, where Model 2 climbs towards the gate through the delay and Model 1 does not. The panels are means: the cartoon's clean separation between conditions survives as an ordering, and the band shows how little of it a single trial sees.

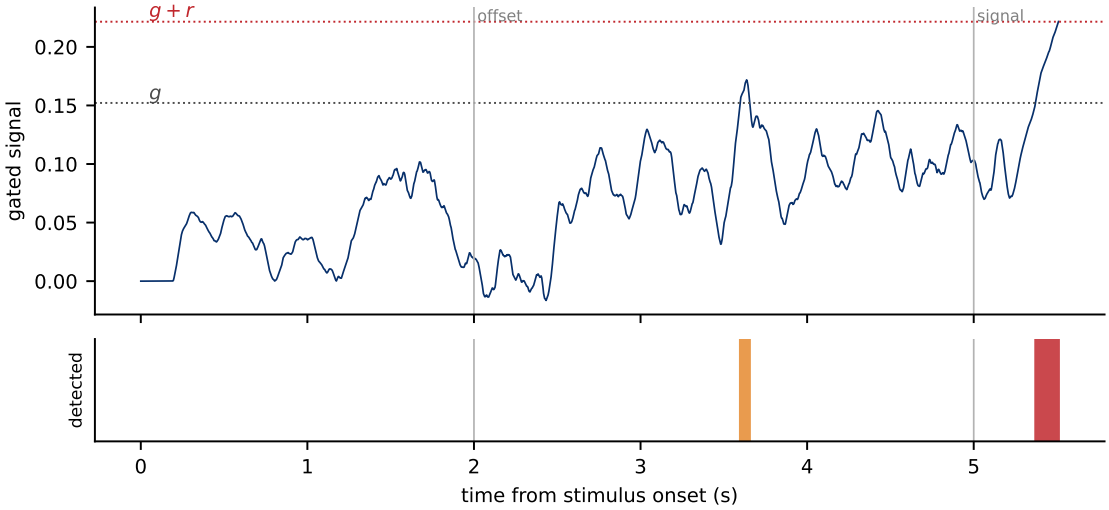


Figure 2: One simulated trial, the counterpart of the manuscript's Figure 2B–C. Above, the gated signal with the gate g and the response criterion $g + r$; below, the intervals the detection rules return, after merging at 25 ms and discarding intervals with too few supra-threshold samples. Orange is a partial activation, red the activation that reaches the criterion and becomes the response.

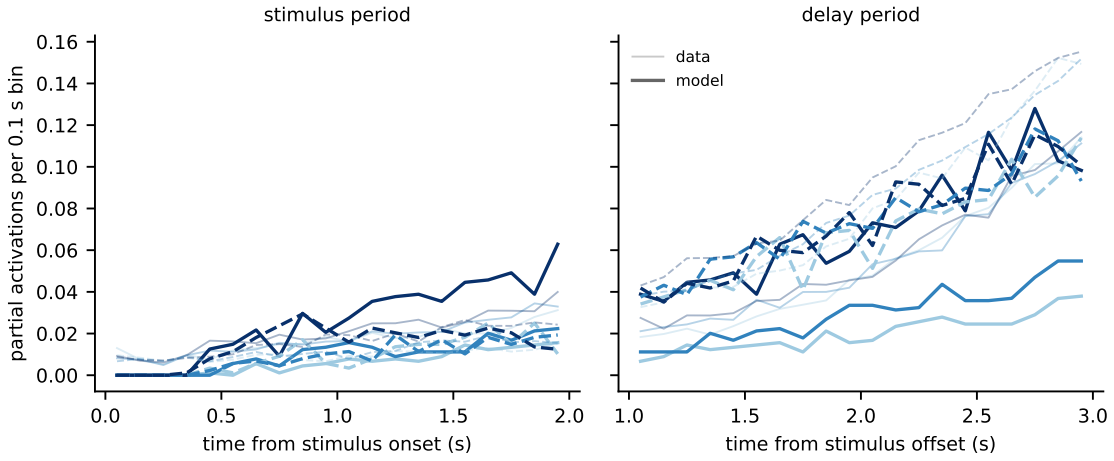


Figure 3: Partial-activation rate, data (thin, pale) against model (thick), at the calibrated parameters with the detection rules applied to the simulated signal. The delay panel shows the main residual: the model spreads the three coherence levels far too widely after a short stimulus, where the data are nearly flat. That is the saturation argument of Section 2 made numerical.

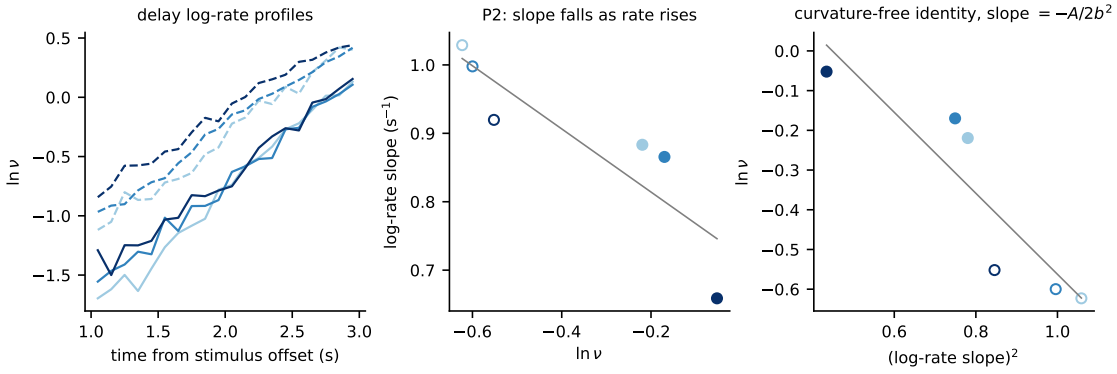


Figure 4: The shape prediction, on data alone. Left, the six offset-locked log-rate profiles. Right, the log-rate slope falls as the cell's rate rises, as (2) requires, since a higher rate means a smaller distance to the gate. Open symbols are the short duration, filled the long.

true ones: where they differ from those published, the difference is flagged and explained in the note attached to the table.

On the window names. The four windows are locked to the four trial events and do not coincide with the intervals they are named after. The onset-locked window runs from stimulus onset to two seconds, so after a 0.3 s stimulus it contains 0.3 s of stimulus and 1.7 s of delay. That is not a detail — it is why the duration effects in that window separate the two models — and it is why this note calls the windows *onset-locked*, *offset-locked*, *signal-locked* and *response-locked* rather than the stimulus, delay, RS and response periods. We suggest the manuscript do the same.

Table S1. Accuracy, response time, premotor and motor execution time

	effect tested	df1	df2	F	p	η_p^2
Accuracy						
R1	duration	1	21	22.68	<.001	0.519
R2	coherence	1.31	27.4	222.22	<.001	0.914
R3	duration × coherence	2	42	12.17	<.001	0.367
Mean response time						
R4	duration	1	21	65.79	<.001	0.758
R5	coherence	1.50	31.39	14.19	<.001	0.403
R6	duration × coherence	2	42	4.61	.016	0.180
Mean premotor execution time						
R7	duration	1	21	69.75	<.001	0.769
R8	coherence	1.59	33.35	14.11	<.001	0.402
R9	duration × coherence	2	42	3.62	.035	0.147
Mean motor execution time						
R10	duration	1	21	1.03	.320	—
	coherence	1.51	31.68	0.54	.537	—
	duration × coherence	2	42	2.39	.104	—

Note. Values as published; this note did not recompute them, since its own extraction differs from the manuscript's by about a hundred trials and substituting it would confuse a difference of analysis with a difference of data. R10 labels the three motor-execution-time rows together: the manuscript states them as one result, that no factor affects motor execution time, and the note treats them as one.

Table S2. Rate of partial activations, responding channel

	effect tested	df1	df2	F	p	η_p^2
Onset-locked window, 0 to 2 s from stimulus onset						
R11	duration	1	21	13.38	.001	0.389
R12	coherence	1.33	28.01	5.25	.021	0.200
R13	time	2.70	56.75	18.05	<.001	0.462
	duration × coherence	2	42	0.312	.734	—
R14	duration × time	5.09	106.82	4.645	<.001	0.181
R15	coherence × time	38	798	1.718	.005	0.076
	duration × coherence × time	38	798	1.058	.377	—
Offset-locked window, 1 to 3 s from stimulus offset						
R16	duration	1	21	33.19	<.001	0.612
R17	coherence	2	42	3.36	.044	0.138
R18	time	1.21	25.43	31.42	<.001	0.599
R19	duration × coherence	2	42	3.474	.040	0.142
R20	duration × time	5.38	112.99	4.520	<.001	0.177
R21	coherence × time	38	798	0.737	.879	—
	duration × coherence × time	38	798	0.959	.542	—
Signal-locked window, 0 to 0.2 s from the response signal						
R22	duration	1	21	36.56	<.001	0.635
R23	coherence	1.56	32.82	1.159	.315	—
R24	time	1	21	21.96	<.001	0.511
	duration × coherence	2	42	0.041	.959	—
R25	duration × time	1	21	7.046	.015	0.251
	coherence × time	2	42	0.407	.668	—
	duration × coherence × time	2	42	0.587	.561	—
Response-locked window, −0.2 to 0 s before the press						
R26	duration	1	21	**7.613**	**.012**	**0.266**
	coherence	1.90	39.98	**1.845**	**.173**	—
	duration × coherence	1.86	39.10	**0.342**	**.697**	—

Note. Values reproduce those published exactly, to the last reported digit, except in the response-locked window, where they are printed in bold because they differ. Two corrected data files were supplied after the note was begun. The signal-locked window did not reproduce with the earlier file and does with the corrected one; nothing there changes. The response-locked window changes both in value and in shape. In the corrected file the bin from −0.1 to 0 s is exactly zero for every participant and cell — a partial activation precedes the response burst by definition, and the burst begins about 160 ms before the press — so the window carries one informative bin, the level and trend scores are the same number up to a constant, and the four time-related rows of the published table are not separate tests. The three effects above are what the window supports. The main effect of duration passes from $F = 0.011$, $p = .919$ as published to $F = 7.613$, $p = .012$; Section 6 sets out the consequences.

One result outside the tables

R27, the levelling off of the rate shortly before the response signal in the two-second, fifty-per-cent condition. It is a reading of Figure 4A rather than a row of any table, and the manuscript discusses it at length, proposing a strategic reduction of temporal preparation to account for it. It is labelled here because Section 2 derives it without that mechanism.

B The rate of partial activations

Throughout, φ and Φ are the standard normal density and distribution, and $\rho := t - \tau$ is elapsed maintenance time.

Define $V(t)$ on $[\tau, t_{\mathrm{RS}})$ by $\mathrm{d}V(t) = \lambda(\varepsilon(t) - V(t))\mathrm{d}t$ with $V(\tau) = 0$, so that $y(t) = m + V(t)$. Since ε admits a continuous modification and V is then a Riemann integral of a continuous function, almost surely $V \in C^1$, and

$$V(\tau + \rho) = \int_0^\rho K(\rho - p)\,\mathrm{d}B(p), \qquad K(v) = \frac{\lambda\sigma_\varepsilon}{\lambda - \kappa}\big(e^{-\kappa v} - e^{-\lambda v}\big),$$

where B drives ε, and $\varepsilon(\tau) = 0$ leaves no additional deterministic term. Note $K(0) = 0$, which is what makes V differentiable, and $K'(0) = \lambda\sigma_\varepsilon$; in the degenerate case $\kappa = \lambda$, $K(v) = \lambda\sigma_\varepsilon\, v\, e^{-\lambda v}$.

With $c := \lambda\sigma_\varepsilon/(\lambda - \kappa)$, standard computations give

$$\begin{aligned}
\operatorname{Var} V(\rho) &= c^2\left[\frac{1 - e^{-2\kappa\rho}}{2\kappa} - \frac{2\big(1 - e^{-(\lambda+\kappa)\rho}\big)}{\lambda + \kappa} + \frac{1 - e^{-2\lambda\rho}}{2\lambda}\right],\\
\operatorname{Var} \dot V(\rho) &= c^2\left[\kappa\,\frac{1 - e^{-2\kappa\rho}}{2} - \frac{2\kappa\lambda\big(1 - e^{-(\lambda+\kappa)\rho}\big)}{\lambda + \kappa} + \lambda\,\frac{1 - e^{-2\lambda\rho}}{2}\right],\\
\operatorname{Cov}\big(V(\rho), \dot V(\rho)\big) &= \tfrac{1}{2}K(\rho)^2.
\end{aligned}$$

As $\rho \to \infty$,

$$\sigma_V^2 = \frac{\lambda\,\mathrm{SD}_\varepsilon^2}{\lambda + \kappa}, \qquad \sigma_{\dot V}^2 = \frac{\lambda^2\kappa\,\mathrm{SD}_\varepsilon^2}{\lambda + \kappa}, \qquad \mathrm{Cov} = 0, \qquad \omega := \frac{\sigma_{\dot V}}{\sigma_V} = \sqrt{\lambda\kappa}. \tag{4}$$

The preparation ramp is added outside the filter, so it shifts the mean and leaves σ_V and ω untouched; (4) holds for both models.

On $[\tau, t_{\mathrm{RS}})$, since $m = s_{\mathrm{resp}}|m|$, we have $s_{\mathrm{resp}}\tilde y(t) = |m| + s_{\mathrm{resp}}V(t) + \beta_t$ with $\beta_t := \beta(t - t_{\mathrm{offset}})^+/\Delta$, and $s_{\mathrm{resp}}V$ has the same law as V because V starts at zero and is driven by W_ε, independent of W_x. A partial activation on the selected channel therefore begins exactly when $s_{\mathrm{resp}}V$ crosses

$$d(t) := g - |m| - \beta\,\frac{(t - t_{\mathrm{offset}})^+}{\Delta}$$

upward. The non-selected channel activates when $s_{\mathrm{resp}}V$ falls below $-g - |m| - \beta_t$, which at the calibrated parameters is 7.1 standard deviations away against 1.6 for the selected one; it is not considered further. The model therefore predicts the non-selected channel to

Denote $\nu(t)$ the upcrossing rate at time $t > \tau$ and let $Z(t) := V(t) - d(t)$, so that

upcrossings of d by V are upcrossings of zero by Z and $\dot{Z} = \dot{V} + \alpha$. We neglect the trajectories that reach r before the end of the delay, as these anticipations are rare, fewer than 1.5% of trials, and excluded from the empirical analysis. Since $V \in C^1$ and Z is Gaussian and non-degenerate for $\rho > 0$, the Kac–Rice formula gives (Azaïs and Wschebor, 2009, chapter 3)

$$\nu(t) = p_{Z(t)}(0)\,\mathbb{E}\left[\dot{Z}(t)\mathbb{1}_{\{\dot{Z}(t)>0\}} \;\middle|\; Z(t) = 0\right].$$

Since $Z(t) \sim N(-d(t), \sigma_V^2)$ and φ is even, $p_{Z(t)}(0) = \sigma_V^{-1}\varphi(d(t)/\sigma_V)$. Moreover $(Z, \dot{Z})$ is jointly Gaussian, so $\dot{Z}(t) \mid Z(t) = 0 \sim N(M_t, S_t)$ with

$$M_t = \alpha + \frac{\mathrm{Cov}(V, \dot{V})}{\sigma_V^2}\, d(t), \qquad S_t = \sigma_{\dot{V}}^2 - \frac{\mathrm{Cov}(V, \dot{V})^2}{\sigma_V^2},$$

and $\mathbb{E}[X\mathbb{1}_{\{X>0\}}] = M\Phi(M/\sqrt{S}) + \sqrt{S}\,\varphi(M/\sqrt{S})$ for $X \sim N(M, S)$. At stationarity $\mathrm{Cov} = 0$, so $M_t = \alpha$ and $S_t = \sigma_{\dot{V}}^2$, and dividing through by σ_V gives, with $a := d/\sigma_V$ and $b := \alpha/\sigma_V$,

$$\nu(t) = \varphi\big(a(t)\big)\left[b\,\Phi\left(\frac{b}{\omega}\right) + \omega\,\varphi\left(\frac{b}{\omega}\right)\right],$$

which is (1). When $\alpha = 0$ this reduces to the classical Rice formula $\nu = (\omega/2\pi)e^{-a^2/2}$, and in particular $\nu \le \omega/2\pi$, attained when the gate sits at the maintained level.

C Draft text for the missing Methods section

The submitted manuscript has a gap on page 18, between the leakage sensitivity analysis and the electromyographic acquisition, where the Methods for the second simulation stage should be. What follows is draft text for that gap, together with the two additional paragraphs the detection procedure requires. It is written to be pasted and edited, and the numbers in it should be regenerated from the final fit.

Second-stage simulation: calibration to the electromyographic data

The first stage established that both candidate models generate the motivating response-time effects across broad regions of parameter space. Because behaviour alone provided limited leverage, a second stage calibrated each model to the electromyographic and behavioural data jointly and compared their fits.

All parameters were free, with one exception. The model is invariant under a common rescaling of the decision variable, the motor-preparatory state, the internal threshold, gating inhibition, the response criterion and the amplitude of maintenance fluctuations, so the diffusion coefficient of motion-evidence accumulation was fixed at 0.100 to define the amplitude unit. In particular the response criterion and the residual motor latency, which the first stage inherited from immediate-report fits, were estimated here rather than assumed; they were held constant across conditions.

Each model was fitted by minimising a normalised discrepancy over nine families of measurements, each comprising the six coherence by duration cells: response accuracy; the mean and interquartile range of premotor execution time; the mean and interquartile range of motor execution time; the rate of partial muscle activations during the delay period; the slope of the logarithm of that rate; the rate during the onset-locked

window; and the proportion of responses preceding the response signal. Each family was scaled by the between-participant standard error of the corresponding observed value, so that families measured in different units contributed comparably. Dispersion was quantified by the interquartile range rather than the standard deviation because the empirical distribution of premotor execution time has a long right tail, in 2.7% of trials exceeding one second, which the model does not reproduce and which would otherwise dominate the criterion.

Stochastic differential equations were integrated with a 1-ms step, with 250 trials per cell during the search and 900 for the reported fit, and with common random numbers across evaluations so that the criterion was a deterministic function of the parameters. Minimisation used the Nelder–Mead simplex in logarithmic parameter coordinates, started from a closed-form solution obtained from the analytic relations between the observable quantities and the parameters, and restarted with progressively larger initial simplices.

Simulated muscle activity was subjected to the same detection rules as the recorded signal. Intervals during which the gated drive to the responding effector exceeded zero were extracted; intervals separated by 25 ms or less were merged; merged intervals containing fewer than three supra-threshold samples were discarded; and one partial muscle activation was counted per surviving interval, except for the interval leading to the response. Burst durations were not fitted, because the onset-refinement step of the detection procedure displaces the reported boundaries by an unknown amount.

Model 1 was fitted independently rather than evaluated at the parameters obtained for Model 2, so that the comparison would not confound the shape of the offset-locked dynamics with their overall level.

Robustness was assessed by multiplying each free parameter in turn by 0.5 and by 2, holding the others at their fitted values, and re-evaluating every directional prediction; and by repeating the fit with a leak common to both accumulators at 0.25, 0.50 and 1.00 s^{-1}.

Additions to the electromyographic Methods

Muscle activation onsets were detected with the `myonset` *package. The rectified signal was thresholded at seven standard deviations above the baseline mean; intervals of supra-threshold activity separated by less than 0.025 s were merged; merged intervals containing fewer than three supra-threshold samples were discarded; and the boundaries of each remaining interval were refined using the integrated-profile algorithm. A partial muscle activation was defined as any detected interval other than the one immediately preceding the button press.*

Two further sentences are worth adding, and are drafted here in ordinary prose rather than as manuscript text, because whether to include them is an editorial decision. The refinement step can only move interval boundaries earlier, so the final event list contains activations separated by less than the merge interval; this does not affect the counts, since the refinement does not change the number of intervals. And 18.5% of trials contain an activation ending within 25 ms of the response burst, so response-period effects should be interpreted with the possibility of segmentation in mind.

D Software

Two languages, split by job. Python owns the model: the simulation, the detector, the calibration, the sweeps, the extraction of events from the archive, and the figures. R owns the statistics: every analysis of variance, every trend decomposition, and the Poisson analyses. They meet at two files written by `export_for_R.py`, holding counts and exposures for the data and for each fitted model, so that the identical statistical analysis runs on both sides.

The statistics exist in two interchangeable versions. `run_stats.R` uses base R only, with the repeated-measures machinery implemented in `rmanova.R`; it is what produced the numbers in this note, and it exists because the analysis container cannot reach CRAN. `run_stats_pkg.R` is the same analysis written with `afex`, `emmeans` and `lme4`, to be run on an ordinary machine; the values the first version produced are listed at the end of it for comparison. The two should agree except in the mixed-model coefficients, where a difference of a few per cent is expected because the two-stage estimator weights participants equally and the mixed model weights them by precision.